

## ANSWER KEY

HMM02-A

1. b) 0
2. c)  $\begin{bmatrix} \frac{1}{5} & \frac{2}{5} \\ \frac{2}{5} & -\frac{1}{5} \end{bmatrix}$
3. a)  $\begin{bmatrix} -191 & -110 \\ 77 & 44 \end{bmatrix}$
4. b) (i) & (iii) only
5. a) (0,0,0), (0,0,1), (0,1,0), (0,1,1), (1,0,0), (1,0,1), (1,1,0), (1,1,1)}
6. c)  $(-\infty, 2)$
7. b) 25
8. b)  $\frac{3}{5}$
9. b) 4
10. c) 10
11. d) 125
12. c) 55
13. a) 1 or 6
14. a) 10080
15. d) xyz
16. d)  $x = 0, y = 0$
17. c) 9
18. b) -1
19. b) -2
20. b) 1
21. c)  $\frac{1}{2}$
22. d)  $\phi$
23. a)  $x = 1$
24. b) (3, 2)
25. Cancelled
26. b)  $x$
27. c) 13
28. c)  $\frac{1}{8}$
29. b)  $\frac{\sin 2^n A}{2^n \sin A}$
30. a)  $\frac{\pi}{4}$
31. d)  $\pi(\sqrt{2}-1)$
32. d)  $\frac{1}{2} - \frac{\pi}{4\sqrt{3}}$
33. a) 0
34. b) 0
35. b)  $(0, \pi)$
36. c)  $\cot^{-1}(a^2)$
37. c)  $4 - \pi$
38. a)  $\log(2\sqrt{2}) + \frac{\pi}{12}$
39. a)  $\pi$
40. a) 0
41. c) 1 & 1
42. a) 2 & 1
43. d) 4
44. d) 0
45. d)  $\frac{4}{5}\hat{i} + \frac{3}{5}\hat{j}$
46. d)  $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}\right)$
47. b) 3
48. a)  $\frac{-1}{\sqrt{6}}\hat{i} + \frac{2}{\sqrt{6}}\hat{j} - \frac{1}{\sqrt{6}}\hat{k}$
49. d)  $\frac{1}{2}|\vec{a} - \vec{b}|$
50. a)  $5\sqrt{3}$  ച. യൂണിറ്റ്
51. b) 0.68
52. a) 11540
53. c) രണ്ട് മാത്രം
54. b)  $\frac{4}{5}$
55. d) -0.43
56. d) 38.4
57. c)  $\frac{3}{8}$
58. a) 1
59. c) 0.57
60. d)  $\frac{1}{2}$
61. d)  $\frac{1}{2}$
62. b)  $\frac{4}{3}$
63. b)  $\frac{5}{16}$
64. c)  $36 C_1 \frac{5^{35}}{6^{36}}$
65. b) 2
66. a)  $e^{-3.4}$
67. a)  $\frac{e^{0.4} \times 0.032}{3}$
68. b)  $e^{-2}$
69. c)  $\frac{128}{15}e^{-4}$
70. c) ii ഉം iii ഉം ശരി
71. b) 0.4772
72. d)  $\frac{5}{12}$
73. c) 3
74. b) 10000
75. b) 16
76. a) അസാധ്യ പരികൽപന തള്ളിക്കളയപ്പെടുന്ന ബാല
77. a)  $\frac{3}{8}$
78. b)  $\frac{5}{9}$
79. c) ജനനമരണ നിരക്കുകളുടെ പഠനം
80. c) 137.5
81. c) Nuthalapati Venkata Ramana
82. a) V.T Bhattathirippad-yogakhemasabha
83. b) INA
84. c) Bipin chandra pal
85. d) Sister Nivedita is a desciple of Swami Dayananda
86. c) Madam Bhikaji Kama
87. a) Sreerama krishna Paramahamsa
88. a) Mikhli Mishustin
89. c) NVS02
90. c) Learning for people, planet, prosperity, and peace
91. d) Subianto
92. b) Kerala
93. c) Yaswath Sinha
94. b) Delhi
95. b) Bhupender Yadav
96. d) Arora
97. c) Piaget
98. a) Quiz
99. c) Stimulus variation
100. d) Assimilation